

Chosen Model and Architecture:

I used BERT (Bidirectional Encoder Representations from Transformers) for binary classification of text as either cyberbullying or non-cyberbullying. The model is pre trained and has a fully connected linear layer. It was fine-tuned using labeled data from social media.

Data Preprocessing and Feature Engineering:

Training Dataset:

- Original Labels:
 - 0: Hate Speech
 - 1: Offensive Language
 - 2: Neither
- Label 1 if class is {0,1} (cyberbullying)
- Label 0 if class = {2} (non-cyberbullying)
- Preprocessing Steps:
 - Converted all text to string
 - Used the BERT Tokenizer to truncate/padd to max length of 128

Training Methodology:

Tokenizer: BertTokenizer.from_pretrained('bert-base-uncased')

Model: BertForSequenceClassification with num_labels=2

Split: 80% training, 20% validation (from Davidson dataset)

Batch size: 16 (train), 32 (eval)

Epochs: 3

Optimizer: AdamW (default in Hugging Face Trainer)

Trainer: Hugging Face Trainer class with compute_metrics for evaluation

Hardware: Google Colab (with GPU - T4)

Model Performance:

Evaluated using accuracy, recall, precision, and F1 metrics

```
def compute_metrics(pred):
    labels = pred.label_ids
    preds = pred.predictions.argmax(-1)
    precision, recall, f1, _ = precision_recall_fscore_support(labels, preds,
        average='binary')
    acc = accuracy_score(labels, preds)
    return {'accuracy': acc, 'precision': precision, 'recall': recall, 'f1':
        f1}
```

Strengths:

- BERT captures context well, so it is able to detect subtle or indirect bullying without false positives
- As it was trained and tested on different datasets, the model is able to strongly generalize on unseen data

Weaknesses:

- Training is resource-intensive (GPU is required for efficiency)

Insights and Future Improvements:

- Model trained on one dataset performed well on the unseen benchmark test data

Improvements:

- Incorporate context-aware features (user history, conversation thread)
- Explore ensemble models combining BERT with traditional NLP features
- Focus on minimizing resources and improving model's efficiency

Dataset:

```
@inproceedings{hateoffensive,  
  title = {Automated Hate Speech Detection and the Problem of Offensive  
Language},  
  author = {Davidson, Thomas and Warmsley, Dana and Macy, Michael and  
Weber, Ingmar},  
  booktitle = {Proceedings of the 11th International AAI Conference on  
Web and Social Media},  
  series = {ICWSM '17},  
  year = {2017},  
  location = {Montreal, Canada},  
  pages = {512-515}  
}
```